AI Engineer (Data): training plan

Novel Vista x ITC Infotech. 48 hours, 6 weeks, 2 sessions a week of 4 hours. Data engineers with 4+ years. Built on ITC's updated syllabus (June 2026), with every lab hands-on and set in one made-up company, FreshCart. Labs 1 to 4 run on Azure Databricks; from Day 5 the labs run in Microsoft Foundry.

Phase summary

Phase	Days	Sessions	Hours	Focus
0. Set up the environment	Day 0	1	0.5	Setup check (30 min) and ITC-provided pre-work. Every participant proves the lab environment works before Day 1.
1. Build the retrieval pipeline	1 to 4	4	16	Embeddings, chunking, vector stores, hybrid retrieval, reranking, grounding with citations.
2. Tools, NL2SQL and agents	5 to 8	4	16	Function calling, OpenAPI and MCP; NL2SQL with schema grounding and semantic layers; agents in Azure AI Foundry; Copilot agents and Microsoft Agent Framework.
3. Fabric pipelines and MLOps	9 to 10	2	8	Microsoft Fabric pipelines and Data Agents, LLM enrichment, MLOps, drift, cost and agent monitoring.
4. Govern and ship	11 to 12	2	8	PII, RBAC on vector stores, responsible AI and Foundry governance, then the capstone Enterprise Data Agent and panel review.
Programme total (Days 1 to 12)		**12**	**48**	**Target from the syllabus: 12 sessions, 48 hours.**

How each 4-hour session runs

Time	Activity	Notes
0:00 to 0:10	Opening and a chat collection	Second day of each week opens with a one-minute recap of the first.
0:10 to 1:15	Teach block 1: plain definition, then analogy, then a live demo that fails before it works	The failure is the lesson. Shown on the trainer's screen, not described.
1:15 to 1:23	Quick check 1: three questions in chat, answers on the next slide	Answers are discussed before moving on.
1:23 to 1:33	Break	Visible timer; exact return time posted in chat.
1:33 to 2:20	Teach block 2	
2:20 to 2:28	Quick check 2, which doubles as the lab briefing	
2:28 to 3:45	Lab in pairs, in breakout rooms	The trainer visits every room twice. Each pair mixes Python strength with domain strength.
3:45 to 4:00	Debrief and end-of-day pulse	Protected time. The lab never eats the debrief.

Day 1 also carries the 38-minute programme opening, so its lab runs 55 minutes with two formats set as homework.

Targets for the cohort

Target	Measure	How it is measured

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Hands-on	Every participant completes every must-do lab	Lab definition of done, checked in the debrief. Catch-up slot offered if missed.
Measurement habit	Every lab deliverable reports a number and the method behind it	Reviewed at the four weekly checkpoints.
Readiness uplift	End-of-programme concept check at least 70%	Same instrument as the Day 1 readiness survey, so the gain is comparable.
Capstone	Every pair scores at least 60% on the capstone rubric	Panel scoring on Day 12, see 'How we check learning'.
Experience	End-of-day pulse averages at least 4 out of 5	Syllabus requirement: end-of-day pulse and end-of-week retro.

Day-by-day plan: Day 0 plus 12 sessions of 4 hours

Must-do lab = everyone completes it. Extra challenge = for pairs who finish early. Blue row = Day 0. Yellow rows = checkpoint days (end of each week).

Day	Week	Phase	Topic	Goal for the day	What we teach (with live demo)	Must-do lab (everyone)	Extra challenge	Lab data	How we check	Case study (20 to 25 min)	Hours	Syllabus topics covered	Format
D0	Week 0	0. Set up the environment	Setup check	Every participant runs the lab environment once and pastes the output in chat.	No teaching. A 30-minute live check a few days before Day 1.	Run the setup commands and the Lab 1 starter; confirm the verification table prints.		DS1	Output pasted in chat; trainer fixes blockers live		0.5	Pre-work is provided by ITC (data landscape primer, Python/SQL/dbt refresher)	Live online, 30 min
1	Week 1	1. Build the retrieval pipeline	Embeddings and chunking	Turn five messy formats into one indexed corpus, and find what silently broke.	Programme opening and word game. What a language model can and cannot do. Embeddings as addresses in meaning-space. Similar vs relevant vs correct. Chunking strategies, chunk size, overlap, metadata. Loaders.	Lab 1: ingest five file formats into one indexed corpus. Verify extraction before indexing. Find the query where the superseded 2022 policy outranks the current one.	Add OCR for the scanned SOP and show the escalation question now retrieves the right document.	DS1	Quick checks 1 and 2; worst chunk contest in the debrief		4	Embeddings; chunking; LangChain / LlamaIndex data loaders	Live class with trainer
2	Week 1	1. Build the retrieval pipeline	Vector stores, recall and latency	Pick a vector store with numbers an architect would accept.	What a vector database does. Exact vs approximate search. Recall, p50 and p95, build time, memory. HNSW and IVF parameters. Pre-filter vs post-filter. How benchmarks lie.	Lab 2: benchmark three stores on the same vectors, sweep the search parameter, run the selective filter test, write a one-page decision memo.	Scale the corpus 100 times and find where exact search stops meeting a 150 ms p95 target.	DS1	Checkpoint 1: decision memo reviewed; re-vote on 'what breaks first'		4	Vector DBs: benchmarking on recall; latency	Live class with trainer
3	Week 2	1. Build the retrieval pipeline	Hybrid retrieval, reranking, filtering	Fix last week's failed queries and prove the lift with nDCG.	Dense and keyword search fail in opposite directions. Reciprocal rank fusion. Retrieve wide, rerank narrow. nDCG in plain terms. Filters as correctness controls.	Lab 3: add BM25, fuse with RRF, rerank the top 50, add a recency filter. Report nDCG@10 at four stages and one query that got worse.	Tune the fusion constant and the weighting; show whether it beats equal weighting on your judged set.	DS1, DS3	Quick checks; four-stage results table		4	Hybrid retrieval; cross-encoder reranking; metadata filtering	Live class with trainer
4	Week 2	1. Build the retrieval pipeline	Grounding, citations, refusal	Ship a RAG API whose citations are verified in code and which refuses when it should.	Why models invent: three causes, three fixes. The grounding instruction. What a citation must be. Faithfulness, answer relevance, context relevance. Evaluation sets with unanswerable questions. The response contract.	Lab 4: RAG API over the Day 3 stack, with verified citations, a refusal threshold and all five response fields. Report refusal rates on answerable and unanswerable questions separately.	Handle conflicting sources: when both policies retrieve, prefer the current one and say so.	DS1, DS3	Checkpoint 2: phantom citations caught; two refusal rates reported	C1	4	Grounding; RAG with citations; hallucination control	Live class with trainer
5	Week 3	2. Tools, NL2SQL and agents	Function calling, OpenAPI and MCP	Give an agent tools it calls safely, and serve one tool two ways.	Function calling and what the model sees of a tool. Three lanes kept in code. Orchestration patterns: one call, a chain, parallel, a loop. OpenAPI files as tools. MCP servers and how the three layers stack. Azure AI Foundry tour: model catalog, playground vs API, deployments.	Lab 5 in Foundry: ground an agent on the refund policy, add the Microsoft Learn MCP server, deploy get_metric to an Azure Function App, connect it as an OpenAPI tool and as an MCP server, then hand the answer to a second agent.	Give two tools the same function name, watch the agent pick the wrong one, and fix it with names and descriptions only.	DS1, DS2	Quick checks; the same March revenue shown through both tools		4	Function calling fundamentals; tool orchestration patterns; OpenAPI integration; MCP server concepts; Azure AI Foundry tour (from Day 1)	Live class with trainer
6	Week 3	2. Tools, NL2SQL and agents	NL2SQL, semantic layers and insight narration	Make an agent's SQL defensible, then move the definitions into a semantic layer.	NL2SQL and six ways a query is wrong without an error. Schema, business rules and worked examples. Three gates on generated SQL. Execution accuracy against reviewed answers. Semantic layers. Narration: number, comparison, scope, never cause. Chart rules and charts that mislead.	Lab 6 in Foundry: an agent writes its own SQL on a small Azure SQL warehouse; five questions asked with the schema only, with business rules and with worked examples, each checked against a reviewed answer key; then a semantic layer agent that also draws charts.	Ask the semantic layer agent for a chart that would mislead, such as a count without its base, and see whether its instructions stop it.	DS5	Checkpoint 3: the five-by-three results table, and one wrong number traced to its rows		4	NL2SQL pipelines with schema grounding; few-shot patterns; semantic layers; insight narration and chart generation	Live class with trainer
7	Week 4	2. Tools, NL2SQL and agents	Agents in Azure AI Foundry	Build an agent that plans several steps over the Lab 5 and Lab 6 tools and keeps context.	Agents vs pipelines. Planner and executor designs. Memory systems. The Agent Skills Framework. Agents in Azure AI Foundry: orchestration, multi-step reasoning, memory and context handling.	Lab 7 in Foundry: a basic enterprise agent that plans several steps over the Lab 5 and Lab 6 tools.	Shared before the session	DS1, DS2, DS5	Quick checks; one agent trace reviewed in the debrief		4	Agents vs pipelines; Planner-Executor architectures and memory systems; Agent Skills Framework; agents in Azure AI Foundry	Live class with trainer
8	Week 4	2. Tools, NL2SQL and agents	Copilot agents and Microsoft Agent Framework	Build a data agent that calls tools over enterprise data, inside the Microsoft stack.	Copilot agents. Microsoft Agent Framework. M365 Copilot Agent Toolkit SDK. Tool-driven agents connecting APIs and enterprise data. Multi-tool workflows and orchestrating decision flows.	Lab 8: an AI data agent using Foundry and tool calling over enterprise data.	Shared before the session	DS2, DS5	Checkpoint 4: the agent's tool calls traced for one decision	C3	4	Copilot agents and Microsoft Agent Framework; M365 Copilot Agent Toolkit SDK; tool-driven agents; multi-tool workflows	Live class with trainer
9	Week 5	3. Fabric pipelines and MLOps	Fabric pipelines and Data Agents	A governed pipeline that re-indexes what changed and enriches records with a model.	Microsoft Fabric fundamentals. Azure Data Factory and Fabric pipelines. Data ingestion for GenAI. Fabric IQ, Fabric Data Agents and Work IQ. Integrating Fabric with Foundry. Data quality, lineage and LLM enrichment in pipelines.	Lab 9: a governed ingestion and re-index pipeline feeding the AI system; LLM-enrich records at scale.	Shared before the session	DS1, DS2	Quick checks; one pipeline run shown with its lineage	C2	4	Microsoft Fabric fundamentals; Azure Data Factory and Fabric pipelines; Fabric IQ, Fabric Data Agents and Work IQ; data quality, lineage and LLM enrichment	Live class with trainer
10	Week 5	3. Fabric pipelines and MLOps	MLOps and agent monitoring	Know within a day when quality, drift or cost has moved, and block a bad deploy.	MLOps for GenAI: MLflow, CI/CD, evaluation harness. Data, embedding and prompt drift. Cost monitoring. Agent monitoring: Foundry evaluation dashboards, experiment tracking, prompt and decision traceability.	Lab 10: an agent monitoring dashboard, a CI gate on evaluation regressions, and drift and cost tracking per pipeline.	Shared before the session	DS3, DS4	Checkpoint 5: which signal caught which fault	C4	4	MLOps for GenAI: MLflow, CI/CD, eval harness; data, embedding and prompt drift; cost monitoring; agent monitoring in Foundry	Live class with trainer
11	Week 6	4. Govern and ship	Governance, RBAC and responsible AI	Two users ask the same question and correctly get different answers, with an audit record for each.	PII handling, compliance and data residency. RBAC on vector stores and knowledge-base security. Responsible AI. Foundry governance: role-based access, model governance, audit logging. Capstone kick-off.	Lab 11: PII detection and RBAC or row-level security on the vector store; audit logging in Foundry; start the capstone build.	Shared before the session	DS1, DS2	Quick checks; no restricted chunk in either context window	C5	4	PII, compliance and data residency; RBAC on vector DBs and knowledge-base security; responsible AI; Foundry governance and audit logging; capstone kick-off	Live class with trainer
12	Week 6	4. Govern and ship	Capstone: Enterprise Data Agent	Show a working, measured, governed agent and defend it.	Short opening, then build. Panel asks three published questions: How do you know? Show me what was retrieved. What would you fix first?	Capstone: an AI Foundry-powered Enterprise Data Agent: ingestion, vector search on Azure AI Search, RAG, agent layer with tool calling, NL2SQL, monitored, evaluated and RBAC-governed.		All	Capstone rubric; end-of-programme concept check	C6	4	Capstone build completion; presentation; trainer panel review	Live class with trainer

Sessions (Days 1 to 12)	**12**
Programme hours (Days 1 to 12)	**48**
Day 0 hours	**0.5**

Lab catalogue (12 labs, Days 1 to 12)

Each lab is about 50 to 85 minutes of must-do work. Labs 1 to 4 run on Azure Databricks; Labs 5 to 12 run in Microsoft Foundry and the Microsoft stack. Details for Labs 7 to 12 are shared before each session. What to upload as proof is what you hand in on the ITC portal.

Lab	Day	Lab name	What people get	What they do	What they hand in	How it is checked	Pass mark	Extra challenge	Builds on	Used later in	What to upload as proof
L01	1	Ingest five formats, and compare three loaders	Starter file; seven-file corpus with planted defects; answer template	Load, verify extraction before indexing, chunk per document type, attach metadata, embed, index, run five queries. Then load the same seven files with the LlamaIndex readers and with the LangChain loaders, and compare unit counts, characters and metadata against your own.	Answer template: empty source, chunking choices, worst chunk, superseded-policy query. Plus a three column table, your loader against each framework, and which one you would ship.	Debrief review against definition of done	Empty source identified; superseded-policy query found and fixed with the recency filter; all three loaders run over all seven files and the silently empty file is found in each.	OCR the scanned SOP	Day 0	L02, L03, L09	Notebook export with outputs; handouts/lab1_answers.md. Must show 7 files, 17 units, 30 chunks and one EMPTY row.
L02	2	Benchmark three stores	Day 1 corpus; three stores (flat, IVF, SQLite); benchmark harness	Ground truth from exact search; recall, p50, p95, build time, memory; parameter sweep; pre vs post filter. Then a 20 minute discussion, not a build: map the twelve labs onto their Microsoft Fabric and Synapse equivalents and mark the three with no clean equivalent.	One-page decision memo with the curve, and the marked Fabric map.	Checkpoint 1 memo review	All six metrics for all three stores; elbow identified; filter difference explained; three Fabric gaps named.	Find where exact search fails a 150 ms p95	L01	L03	Notebook export; handouts/lab2_memo.md; out/lab2_curve.png. Must show the three-store table and the n_probe curve.
L03	3	Hybrid retrieval and lift	Retrieval module (BM25, RRF, reranker, nDCG); Day 1 corpus	Add keyword search, fuse, rerank, filter. Judge 20 queries. Measure nDCG at four stages.	Four-stage results table; one regression explained	Numbers re-run by trainer	Hybrid at least equal to dense on mean nDCG; every stage measured separately	Tune fusion and weighting	L01, L02	L04	Notebook export showing nDCG@10 at all four stages and the lift between them.
L04	4	RAG API with verified citations	RAG module; grounding prompt; Day 3 stack	Endpoint with citations verified in code, refusal threshold, five response fields. 20 answerable and 5 unanswerable questions.	API; faithfulness score; two refusal rates	Checkpoint 2 practical test	Zero unverified citations surfaced; refuses at least 4 of 5 unanswerable; refuses no more than 10% of answerable	Conflict handling between policy versions	L03	L08, L10, L12	Notebook export with the debrief block: refusal rates, faithfulness, correct facts and phantom citations caught.
L05	5	An agent with tools in Foundry	Lab 5 handbook; get_metric Function App code; OpenAPI file; refund policy PDF; FreshCart warehouse file	Ground an agent on the refund policy; add the Microsoft Learn MCP server; deploy get_metric to an Azure Function App; connect it as an OpenAPI tool and as an MCP server; pass the answer to a second agent through a workflow.	The same March revenue through both tools, and the second agent's email	Shown in the playground in the debrief	531,043.03 through both tools; grounded answers cite the policy; the second agent adds no policy detail of its own	Fix a clash between two tools with the same function name, using names and descriptions only	L04	L06, L07	Screenshots of the agent answering through the OpenAPI tool and the MCP tool, and of the second agent's email.
L06	6	Questions in English, numbers you can defend	Lab 6 handbook; shared Azure SQL warehouse freshcart_mini; a Function App with run_sql and official_number; tool files; four sets of agent instructions; answer key	Build SqlAgent and ask five questions with the schema only, then with business rules, then with worked examples, checking each against the reviewed answer key in the Query editor; then build MetricsAgent with the semantic layer and Code Interpreter for charts.	Results table for five questions at three stages; one wrong number traced to its rows	Checkpoint 3 review	Every cell of the results table filled and checked against the answer key; at least one wrong number traced to its rows	Ask for a misleading chart and see whether the instructions stop it	L05	L07, L08, L12	Screenshot of the filled results table and of one chart drawn by MetricsAgent.
L07	7	A first enterprise agent in Foundry	Shared before the session	Build a basic enterprise agent in Azure AI Foundry that plans several steps over the Lab 5 and Lab 6 tools, with memory.	Shared before the session	Shared before the session	Shared before the session		L05, L06	L08, L12	Shared before the session
L08	8	An AI data agent with Copilot and Agent Framework	Shared before the session	Build an AI data agent with Foundry and tool calling over enterprise data, using Microsoft Agent Framework.	Shared before the session	Shared before the session	Shared before the session		L07	L12	Shared before the session
L09	9	A Fabric pipeline that re-indexes and enriches	Shared before the session	Build a governed ingestion and re-index pipeline feeding the AI system, and LLM-enrich records at scale.	Shared before the session	Shared before the session	Shared before the session		L01, L03	L10, L12	Shared before the session
L10	10	Agent monitoring and CI gates	Shared before the session	Build an agent monitoring dashboard, a CI gate on evaluation regressions, and drift and cost tracking per pipeline.	Shared before the session	Shared before the session	Shared before the session		L04, L09	L12	Shared before the session
L11	11	PII, RBAC and audit in Foundry	Shared before the session	PII detection, RBAC or row-level security on the vector store, and audit logging in Foundry; start the capstone build.	Shared before the session	Shared before the session	Shared before the session		L03, L04	L12	Shared before the session
L12	12	Capstone: Enterprise Data Agent	Shared before the session	Ingestion, vector search on Azure AI Search, RAG, an agent layer with tool calling, NL2SQL, monitoring and evaluation, and RBAC.	Shared before the session	Panel rubric	At least 60% on the capstone rubric		All		Shared before the session

Notebook export means File, then Export, then IPython notebook, with Include outputs ticked.

Between-session assignments (3)

Individual, 2 to 3 hours, done in the 2 to 3 day gap between sessions that the syllabus builds in for homework. A1 runs in notebook cells on Databricks. A2 extends Lab 6 in Microsoft Foundry and the Azure Query editor. A3 builds on the Day 10 monitoring lab.

A1: Make the corpus answer the escalation question

Given on	Day 2
Due on	Day 3
Time (hours)	2 to 3
Scenario	Support agents say questions about when to escalate get confident wrong answers.
Task	Find why (the SOP is scanned and extracted nothing). Fix it with apply_ocr, which works offline now that the pack ships the recognised text in data/cache. Add an assertion that would catch it next time, then re-run verification and the five Lab 1 queries.
What to hand in	Notebook export with outputs; verification table before and after; one-page note
Extension (optional)	Build a 20-question golden set with 5 unanswerable questions. It seeds Lab 4.
Builds on labs	L01, L02

Scoring	Criterion	Weight (%)
	Root cause identified correctly	25
	Fix verified by an assertion, not by eye	25
	Before and after measured	30
	Note is clear and short	20
	Total	**100**

A2: Extend the Lab 6 semantic layer to new questions

Given on	Day 6
Due on	Day 7
Time (hours)	2 to 3
Scenario	Operations sends ten business questions. Some the semantic layer can answer through a breakdown, some it cannot, and one is ambiguous.
Task	Write 10 new questions on the small warehouse, with reviewed SQL run in the Query editor. Ask SqlAgent (rules and examples) and MetricsAgent all 10, in new chats. Then define one number the layer refused the way the layer stores it: formula, tables, business rules, date column, and its grain in one sentence. Run its SQL, and write the instruction line MetricsAgent would need for it.
What to hand in	Answer key of 10 queries with their numbers; results table for both agents (SqlAgent right; MetricsAgent answered, right, refused); the new number's definition and result; one-page note

Extension (optional)	Write the misleading-chart question for MetricsAgent and report whether its instructions stop it.
Builds on labs	L06

Scoring	Criterion	Weight (%)
	New number declares joins and grain correctly	30
	Test questions cover edge cases	20
	Both agents measured, not claimed	30
	Note is clear and honest	20
	Total	**100**

A3: The silent regression drill

Given on	Day 10
Due on	Day 11
Time (hours)	3
Scenario	After a chunking change, support says answers got worse. Nothing alerted.
Task	Reproduce a silent regression, find which signal moved, write a runbook entry and an alert with a named owner, and add the gate that would have blocked it.
What to hand in	Harness output for both runs; alert rule; runbook entry; gate shown failing on the change and passing without it
Extension (optional)	Show the same regression in the score distribution before answer quality moves.
Builds on labs	L09, L10

Scoring	Criterion	Weight (%)
	Regression reproduced	20
	Right signal identified	25
	Gate blocks the change	30
	Runbook and alert are actionable	25
	Total	**100**

Case studies (5)

Each runs about 20 to 25 minutes inside the teaching block: 5-minute story, 10 minutes of pair discussion, 10-minute debrief. Figures below are taken from the linked sources.

Case	Day	Case study	Sector	What happened	Discussion questions	The lesson	Lab it sets up	Source	Link	Note
C1	4	Air Canada held liable for its chatbot	Airline	Feb 2024: a Canadian tribunal ruled Air Canada must honour wrong advice from its website chatbot about claiming a bereavement fare after purchase. Air Canada argued the chatbot was responsible for its own actions; the tribunal rejected this and awarded about CAD 650 plus interest and fees.	1) At what point should the assistant have refused or linked to the policy? 2) At FreshCart, who owns what the support assistant tells a customer?	The company owns every answer. Cite the source and refuse when there is no evidence.	L04	Moffatt v. Air Canada, 2024 BCCRT 149	https://www.canlii.org/en/bc/bccrt/doc/2024/2024bccrt149/2024bccrt149.html	Not a data company, but the clearest ruling on liability for AI answers
C2	9	England's missing 15,841 COVID cases	Public health	Between 25 September and 2 October 2020, 15,841 positive cases were left out of England's daily figures. Lab results arrived as CSV files and were loaded into Excel templates in the old XLS format, capped at 65,536 rows. Each result used several rows, so each template held about 1,400 cases, and rows beyond the cap were dropped without an error. Contact tracing was delayed.	1) Which expectation from today would have caught this on day one? 2) Why did every job report success?	Job success is not data success. A volume check catches a partial load that every other check passes.	L09	The Register, 5 Oct 2020; AP via PBS NewsHour, 5 Oct 2020	https://www.theregister.com/2020/10/05/excel_england_coronavirus_contact_error/	Row-limit explanation is from press reporting; government described it as a file-size limit
C3	8	Knight Capital's 45-minute deployment failure	Trading	1 Aug 2012: new code was deployed to 7 of 8 servers. The eighth still held dormant legacy code, which the new orders triggered. Processing 212 customer orders, the router sent millions of orders and got over 4 million executions in 154 stocks in about 45 minutes. Knight lost over USD 460 million and later paid a USD 12 million SEC penalty.	1) Which gate would have stopped this before the market opened? 2) What is the equivalent of the eighth server in an AI pipeline?	Verify every deployment, remove dead code paths, and have a way to stop fast. Gates exist for the Friday-evening deploy.	L08	SEC order, Release No. 70694, 16 Oct 2013	https://www.sec.gov/files/litigation/admin/2013/34-70694.pdf	Not an AI system; a release-control failure, which is what Day 8 is about
C4	10	Zillow Offers wound down	Real estate	2 Nov 2021: Zillow wound down its home-buying business. The CEO said the unpredictability in forecasting home prices far exceeded what they anticipated. Q3 2021 included a USD 304.4 million inventory write-down, and the wind-down cut about 25% of the workforce.	1) What would you have monitored to see the forecast drifting from reality? 2) What limit would have capped the damage while you investigated?	Models degrade when the world shifts. Watch prediction against outcome, not just the model's own confidence.	L10	Zillow Group Form 8-K and Q3 2021 shareholder letter, 2 Nov 2021	https://www.sec.gov/Archives/edgar/data/1617640/000161764021000085/z-20211102.htm	Zillow also cited capacity, labour and supply constraints; do not present it as a pure model failure
C5	11	Samsung restricts generative AI after a code leak	Electronics	May 2023: Samsung restricted generative AI tools on company devices after staff uploaded sensitive internal source code to ChatGPT. Its memo said data sent to such services is stored externally and hard to retrieve or delete.	1) Where should the boundary have been drawn before access was opened? 2) How is this the same problem as a restricted chunk reaching the context window?	Once data leaves the boundary it cannot be recalled. Design the boundary before opening the door.	L11	Bloomberg, 2 May 2023; CNBC, 2 May 2023	https://www.cnbc.com/2023/05/02/samsung-bans-use-of-ai-like-chatgpt-for-staff-after-misuse-of-chatbot.html	Bloomberg original is paywalled; CNBC link is open

How we check learning

Checks are frequent and practical. No written exams. Items marked from the syllabus are ITC requirements.

Check	When	Format	Why	Done by
Day-1 readiness survey (syllabus)	Day 1, in class, 30 min	Short concept check, same instrument reused at the end	Calibrates starting depth; gives a baseline for the uplift target	Individual
Quick checks	Two per session, before the break and before the lab	Three applied questions in chat, answers on the next slide	Confirms the section landed before building on it; breaks up a 4-hour virtual session	Individual
Must-do lab	Every session	Definition of done in the lab brief	Everyone reaches the same practical level	Pair
Extra challenge	Every session	Shared in the debrief	Keeps strong participants stretched without splitting the group	Pair
Weekly checkpoint	Days 2, 4, 6, 8, 10	Lab deliverable reviewed: memo, report, gate output	Confirms the week's learning before moving on	Pair
Assignment	Given Days 2, 6, 10; due next session	2 to 3 hours, scored with a rubric	Applies the week to a realistic FreshCart problem	Individual
End-of-day pulse (syllabus)	Last 5 minutes of every session	3 questions: clarity, pace, one thing to change	Lets the trainer adjust within the week	Individual
End-of-week retro (syllabus)	Second session of each week	5-minute structured retro	Syllabus requirement; feeds the weekly report	Cohort
Capstone panel	Day 12	10 min present, 10 min questions, rubric below	Shows the whole pipeline works and can be defended	Pair
End-of-programme check	Day 12	Same as the Day 1 survey	Measures individual uplift	Individual

Capstone scoring

Area	What is measured	Weight (%)	Strong looks like	
Works end to end	One question flows through ingest, retrieval, warehouse, entitlements, answer	20	Demonstrated live, not described	
Evidence	Accuracy, faithfulness, retrieval quality, latency, cost from their own runs	25	Numbers with the method stated and limits named	
Retrieval understanding	Shows what was retrieved, with scores and filters	15	Explains why the top chunk ranked first	
Governance	Entitlement demo and one audit record	15	Restricted chunk shown absent from the context window	
Quality gate	One gate shown blocking a deliberate regression	10	Threshold set above measured noise	
Known gaps	What they did not solve and what they would fix first	15	Specific, prioritised, with a proposed fix	
	Total	**100**		

Pre-reads (45 to 60 minutes a week)

Primary sources only. Quick-check questions draw on these. Week 0 pre-work is provided by ITC.

Week	Reading	Link	For
Week 0	ITC-provided pre-work: data landscape primer; Python / SQL / dbt refresher	Provided by ITC	All
Week 1	Lewis et al. (2020), Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks (intro and Figure 1)	https://arxiv.org/abs/2005.11401	All
Week 1	Malkov and Yashunin, Efficient and robust approximate nearest neighbor search using HNSW graphs (section 1 only)	https://arxiv.org/abs/1603.09320	All
Week 2	Cormack, Clarke and Buettcher (2009), Reciprocal Rank Fusion outperforms Condorcet (2 pages)	https://cormack.uwaterloo.ca/cormacksigir09-rrf.pdf	All
Week 2	Anthropic (2024), Introducing contextual retrieval	https://www.anthropic.com/news/contextual-retrieval	All
Week 3	Yu et al. (2018), Spider: a large-scale text-to-SQL dataset (sections 1 and 2)	https://arxiv.org/abs/1809.08887	All
Week 4	Microsoft Foundry Agent Service overview	https://learn.microsoft.com/en-us/azure/foundry/agents/overview	All
Week 4	Microsoft Agent Framework overview	https://learn.microsoft.com/en-us/agent-framework/overview/	All
Week 5	Observability in Microsoft Foundry: evaluation and monitoring	https://learn.microsoft.com/en-us/azure/foundry/concepts/observability	All
Week 6	OWASP Top 10 for LLM applications	https://genai.owasp.org/llm-top-10/	All
Week 6	Digital Personal Data Protection Act, 2023 (MeitY). Skim sections 4 to 8 for obligations; defer legal questions to ITC's DPO	https://www.meity.gov.in/static/uploads/2024/06/2bf1f0e9f04e6fb4f8fef35e82c42aa5.pdf	All
Week 5	Fabric data agent concepts	https://learn.microsoft.com/en-us/fabric/data-science/concept-data-agent	All
Week 3	Model Context Protocol: introduction	https://modelcontextprotocol.io/docs/2026-07-28/getting-started/intro	All

Lab data and tools

All lab data is made up. No ITC data is used.

ID	Data	What it contains	Used in
DS1	FreshCart document corpus	7 files: current and superseded refund policies, two-column partner contract, scanned SOP with no text layer, SLA table in DOCX, QBR deck with out-of-order text boxes, claims spreadsheet with header on row 5	L01 to L04, L09, L11
DS2	FreshCart warehouse	Star schema: deprecated orders table, legacy status column, internal test accounts, fan-out order items, cumulative snapshot	L05 to L07
DS3	Golden question set	Answerable and unanswerable questions with expected answers and supporting chunks; grows from Week 2	L04, L08, L10, L12
DS4	Injected fault pack	Degraded index, loosened prompt, cost regression, plus three prepared changes for L08	L08, L10
DS5	FreshCart mini warehouse (Azure SQL)	Azure SQL database freshcart_mini: 6 tables and 56 rows with the same traps as DS2, small enough to check every answer by hand; shared with the class, read-only	L06, A2

Area	Syllabus stack	Default choice	Offline fallback (lab repo)
Language	Python 3.11+, SQL	Notebooks with shared starter code	Plain Python scripts
Platform	Azure AI Foundry, Microsoft Fabric, Azure Databricks / Synapse	Azure Databricks for Labs 1 to 4; Microsoft Foundry for Labs 5 and 6	Lab handbooks and files on GitHub
Loaders	LangChain, LlamaIndex	Framework loaders on Databricks	Format-specific loaders in freshcart.py
Vector store	Azure AI Search, pgvector	The store ITC uses, plus Databricks Vector Search	Flat, IVF and SQLite stores
Reranker	Cross-encoder	Approved hosted reranker	Lexical stand-in (measured roughly neutral)
Models	Not specified	gpt-4.1-mini deployed in Foundry for Labs 5 and 6; Llama 3.3 70B on Databricks for Labs 1 to 4	Recorded answers in the Databricks labs
Warehouse and semantic layer	Databricks, dbt	Azure SQL warehouse with a semantic layer in an Azure Function (Lab 6)	SQLite warehouse file
Orchestration	Airflow, Azure Data Factory, Fabric pipelines	Fabric pipelines (Lab 9)	Plain Python DAG
Quality and catalog	Unity Catalog	Pipeline expectations and Unity Catalog lineage	Expectations and lineage map in quality.py

MLOps	MLflow, Evidently, Azure AI Foundry evaluation and monitoring	Foundry evaluation and monitoring, MLflow tracking	observe.py harness
PII	Not specified	Pattern plus context rules; Presidio optional	privacy.py
Agents	Copilot agents, Microsoft Agent Framework, M365 Copilot Agent Toolkit, Fabric Data Agents	Foundry agents (Labs 5 to 8)	